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Exp.7 Network Address Translation for IPv4 (NAT)

Objectives

1. Describe NAT operation.
2. Describe the benefits and drawbacks of NAT.
3. Configure static NAT using the CLI.
4. Configure dynamic NAT using the CLI.
5. Configure PAT using the CLI.
6. Use show commands to verify NAT operation.

NAT Operation:

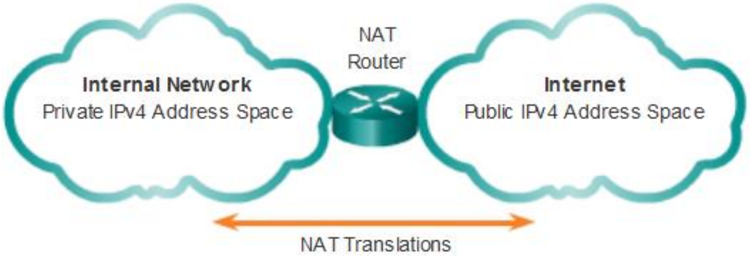
All public IPv4 addresses that transverse the Internet must be registered with a Regional Internet Registry (RIR). Organizations can lease public addresses from a service provider (SP), but only the registered holder of a public Internet address can assign that address to a network device. However, with a theoretical maximum of 4.3 billion addresses (2^{32}), IPv4 address space is severely limited. When Bob Kahn and Vint Cerf first developed the suite of TCP/IP protocols including IPv4 in 1981, they never envisioned what the Internet would become. At the time, the personal computer was mostly a curiosity for hobbyists and the World Wide Web was still more than a decade away. With the proliferation of personal computing and the advent of the World Wide Web, it soon became obvious that 4.3 billion IPv4 addresses would not be enough. The long term solution was IPv6, but more immediate solutions to address exhaustion were required. For the short term, several solutions were implemented by the IETF including Network Address Translation (NAT) and RFC 1918 private IPv4 addresses. This experiment discusses how NAT, combined with the use of private address space, is used to both conserve and more efficiently use IPv4 addresses to provide networks of all sizes access to the Internet.
Networks are commonly implemented using private IPv4 addresses, as defined in RFC 1918. The figure below shows the range of addresses included in RFC 1918.

Private Internet addresses are defined in RFC 1918:

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

These private addresses are used within an organization or site to allow devices to communicate locally. However, because these addresses do not identify any single company or organization, private IPv4 addresses cannot be routed over the Internet. To allow a device with a private IPv4 address to access devices and resources outside of the local network, the private address must first be translated to a public address.
NAT provides the translation of private addresses to public addresses. This allows a device with a private IPv4 address to access resources outside of their private network, such as those found on the Internet. NAT combined with private IPv4 addresses, has proven to be a useful method of preserving public IPv4 addresses. A single, public IPv4 address can be shared by hundreds, even thousands of devices, each configured with a unique private IPv4 address.

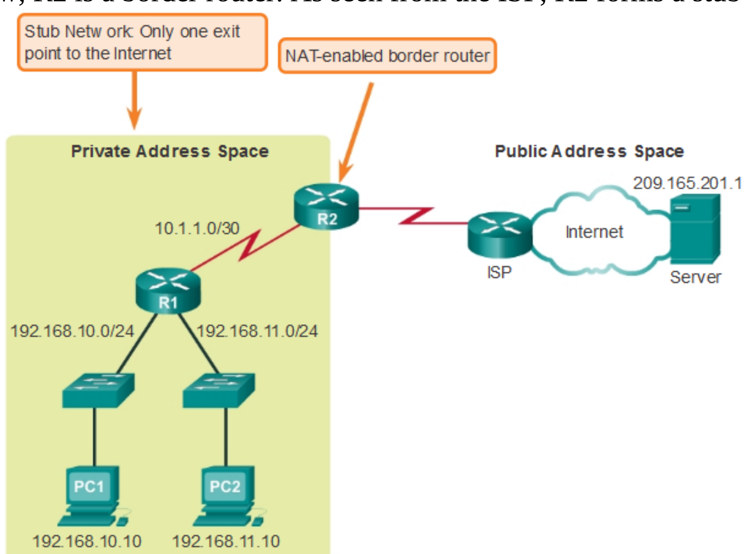
Without NAT, the exhaustion of the IPv4 address space would have occurred well before the year 2000. However, NAT has certain limitations, which will be explored later. The solution to the exhaustion of IPv4 address space and the limitations of NAT is the eventual transition to IPv6.



NAT has many uses, but its primary use is to conserve public IPv4 addresses. NAT has an added benefit of adding a degree of privacy and security to a network, because it hides internal IPv4 addresses from outside networks.

NAT-enabled routers can be configured with one or more valid public IPv4 addresses. These public addresses are known as the NAT pool. When an internal device sends traffic out of the network, the NAT-enabled router translates the internal IPv4 address of the device to a public address from the NAT pool. To outside devices, all traffic entering and exiting the network appears to have a public IPv4 address from the provided pool of addresses.

A NAT router typically operates at the border of a stub network. A stub network is a network that has a single connection to its neighboring network, one way in and one way out of the network. In the example below, R2 is a border router. As seen from the ISP, R2 forms a stub network.



In NAT terminology, the inside network is the set of networks that is subject to translation. The outside network refers to all other networks.

When using NAT, IPv4 addresses have different designations based on whether they are on the private network, or on the public network (Internet), and whether the traffic is incoming or outgoing.

NAT includes four types of addresses:

- Inside local address
- Inside global address
- Outside local address
- Outside global address

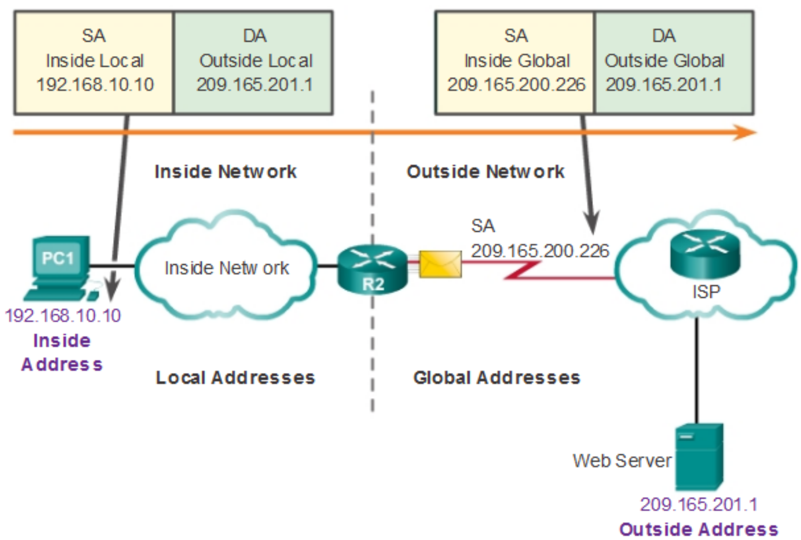
When determining which type of address is used, it is important to remember that NAT terminology is always applied from the perspective of the device with the translated address:

- **Inside address -** The address of the device which is being translated by NAT.
- **Outside address -** The address of the destination device.

NAT also uses the concept of local or global with respect to addresses:

- **Local address -** A local address is any address that appears on the inside portion of the network.
- **Global address -** A global address is any address that appears on the outside portion of the network.

Types of NAT Addresses



Static NAT

Static NAT uses a one-to-one mapping of local and global addresses. These mappings are configured by the network administrator and remain constant.

In the figure, R2 is configured with static mappings for the inside local addresses of Svr1, PC2, and PC3. When these devices send traffic to the Internet, their inside local addresses are translated to the configured inside global addresses. To outside networks, these devices have public IPv4 addresses.

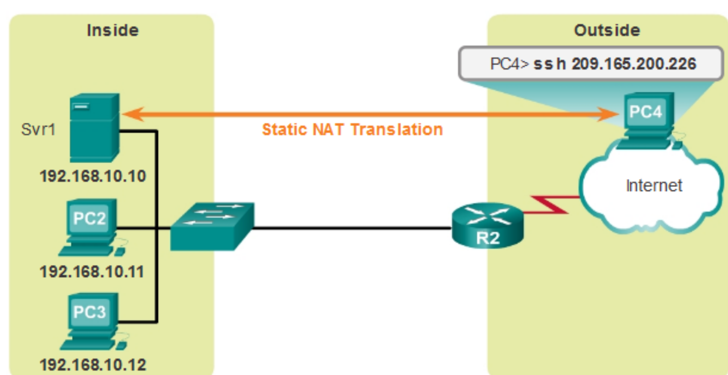
Static NAT is particularly useful for web servers or devices that must have a consistent address that is accessible from the Internet, such as a company web server. It is also useful for devices that must be accessible by authorized personnel when offsite, but not by the general public on the Internet. For example, a network administrator from PC4 can SSH to Svr1's inside global address (209.165.200.226). R2 translates this inside global address to the inside local address and connects the administrator's session to Svr1.

Static NAT requires that enough public addresses are available to satisfy the total number of simultaneous user sessions.

Static NAT

Static NAT Table

Inside Local Address	Inside Global Address - Addresses reachable via R2
192.168.10.10	209.165.200.226
192.168.10.11	209.165.200.227
192.168.10.12	209.165.200.228



Dynamic NAT

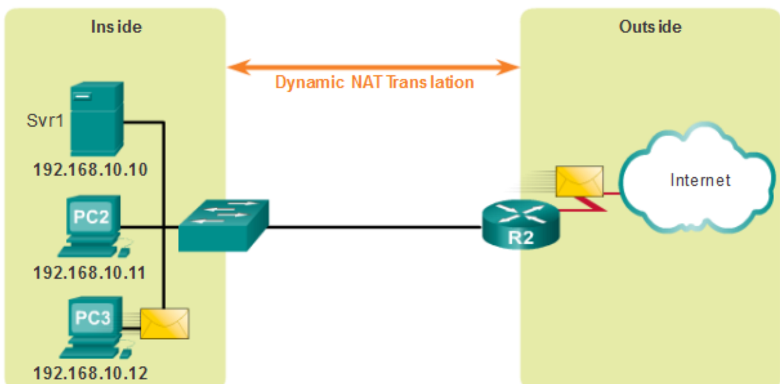
Dynamic NAT uses a pool of public addresses and assigns them on a first-come, first-served basis. When an inside device requests access to an outside network, dynamic NAT assigns an available public IPv4 address from the pool.

In the figure, PC3 has accessed the Internet using the first available address in the dynamic NAT pool. The other addresses are still available for use. Similar to static NAT, dynamic NAT requires that enough public addresses are available to satisfy the total number of simultaneous user sessions.

Dynamic NAT

IPv4 NAT Pool

Inside Local Address	Inside Global Address Pool - Addresses reachable via R2
192.168.10.12	209.165.200.226
Available	209.165.200.227
Available	209.165.200.228
Available	209.165.200.229
Available	209.165.200.230



Port Address Translation (PAT)

Port Address Translation (PAT), also known as NAT overloading, maps multiple private IPv4 addresses to a single public IPv4 address or a few addresses. This is what most home routers do. The ISP assigns one address to the router, yet several members of the household can simultaneously access the Internet. This is the most common form of NAT.

With PAT, multiple addresses can be mapped to one or to a few addresses, because each private address is also tracked by a port number. When a device initiates a TCP/IP session, it generates a TCP or UDP source port value to uniquely identify the session. When the NAT router receives a packet from the client, it uses its source port number to uniquely identify the specific NAT translation.

PAT ensures that devices use a different TCP port number for each session with a server on the Internet. When a response comes back from the server, the source port number, which becomes the destination port number on the return trip, determines to which device the router forwards the packets. The PAT process also validates that the incoming packets were requested, thus adding a degree of security to the session.

PAT

Inside Global Address	Inside Local Address
209.165.200.226:1444	192.168.10.10:1444
209.165.200.226:1445	192.168.10.11:1444
209.165.200.226:1555	192.168.10.12:1555
209.165.200.226:1556	192.168.10.13:1555

Note that PAT attempts to preserve the original source port. However, if the original source port is already used, PAT assigns the first available port number.

Benefits of NAT

NAT provides many benefits, including:

- NAT conserves the legally registered addressing scheme by allowing the privatization of intranets. NAT conserves addresses through application port-level multiplexing. With NAT overload, internal hosts can share a single public IPv4 address for all external communications. In this type of configuration, very few external addresses are required to support many internal hosts.
- NAT increases the flexibility of connections to the public network. Multiple pools, backup pools, and load-balancing pools can be implemented to ensure reliable public network connections.
- NAT provides consistency for internal network addressing schemes. On a network not using private IPv4 addresses and NAT, changing the public IPv4 address scheme requires the readdressing of all hosts on the existing network. The costs of readdressing hosts can be significant. NAT allows the existing private IPv4 address scheme to remain while allowing for easy change to a new public addressing scheme. This means an organization could change ISPs and not need to change any of its inside clients.
- NAT provides network security. Because private networks do not advertise their addresses or internal topology, they remain reasonably secure when used in conjunction with NAT to gain controlled external access. However, NAT does not replace firewalls.

NAT does have some drawbacks. The fact that hosts on the Internet appear to communicate directly with the NAT-enabled device, rather than with the actual host inside the private network, creates a number of issues.

One disadvantage of using NAT is related to network performance, particularly for real time protocols such as VoIP. NAT increases switching delays because the translation of each IPv4 address within the packet headers takes time. The first packet is process-switched; it always goes through the slower path. The router must look at every packet to decide whether it needs translation. The router must alter the IPv4 header, and possibly alter the TCP or UDP header. The IPv4 header checksum, along with the TCP or UDP checksum must be recalculated each time a translation is made. Remaining packets go through the fast-switched path if a cache entry exists; otherwise, they too are delayed.

Another disadvantage of using NAT is that end-to-end addressing is lost. Many Internet protocols and applications depend on end-to-end addressing from the source to the destination. Some applications do not work with NAT. For example, some security applications, such as digital signatures, fail because the source IPv4 address changes before reaching the destination. Applications that use physical addresses, instead of a qualified domain name, do not reach destinations that are translated across the NAT router. Sometimes, this problem can be avoided by implementing static NAT mappings.

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in
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out.

End-to-end IPv4 traceability is also lost. It becomes much more difficult to trace packets that undergo numerous packet address changes over multiple NAT hops, making troubleshooting challenging.

Configuring Static NAT

Static NAT is a one-to-one mapping between an inside address and an outside address. There are two basic tasks when configuring static NAT translations.

Step 1. The first task is to create a mapping between the inside local address and the inside global addresses. For example, the 192.168.10.254 inside local address and the 209.165.201.5 inside global address in the figure below are configured as a static NAT translation.

private public

Router(config)# ***ip nat inside source static*** *local-ip global-ip* for servers

Step 2. After the mapping is configured, the interfaces participating in the translation are configured as inside or outside relative to NAT. In the example, the Serial 0/0/0 interface of R2 is an inside interface and Serial 0/1/0 is an outside interface.

Router(config)# ***interface*** *type number*

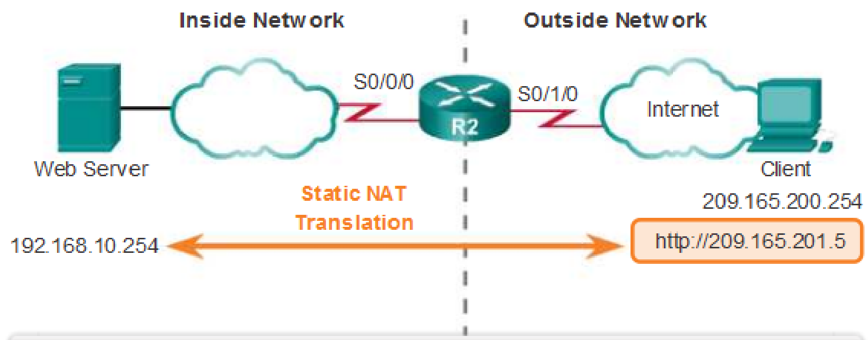
Router(config-if)# ***ip nat {inside | outside}*** → جهة الربط بال ISP.

sh ip nat translation

sh ip nat statistics

The figure below shows an inside network containing a web server with a private IPv4 address. Router R2 is configured with static NAT to allow devices on the outside network (Internet) to access the web server. The client on the outside network accesses the web server using a public IPv4 address. Static NAT translates the public IPv4 address to the private IPv4 address.

Example Static NAT Configuration




```
Establishes static translation between an inside local address and
an inside global address.
R2(config)# ip nat inside source static 192.168.10.254 209.165.201.5

R2(config)# interface Serial0/0/0
R2(config-if)# ip address 10.1.1.2 255.255.255.252
Identifies interface serial 0/0/0 as an inside NAT interface.
R2(config-if)# ip nat inside
R2(config-if)# exit

R2(config)# interface Serial0/1/0
R2(config-if)# ip address 209.165.200.225 255.255.255.224
Identifies interface serial 0/1/0 as the outside NAT interface.
R2(config-if)# ip nat outside
```

A useful command to verify NAT operation is the **show ip nat translations** command. This command shows active NAT translations. Static translations, unlike dynamic translations, are always in the NAT table. The figure below shows the output from this command using the previous configuration example. Because the example is a static NAT configuration, the translation is always present in the NAT table regardless of any active communications. If the command is issued during an active session, the output also indicates the address of the outside device as shown below.

The static translation is always present in the NAT table.

```
R2# show ip nat translations
Pro Inside global  Inside local   Outside local  Outside global
--- 209.165.201.5  192.168.10.254 ---             ---
R2#
```

The static translation during an active session.

```
R2# show ip nat translations
Pro Inside global Inside local   Outside local   Outside global
--- 209.165.201.5 192.168.10.254 209.165.200.254 209.165.200.254
R2#
```

Another useful command is the **show ip nat statistics** command. As shown in the figure below, the **show ip nat statistics** command displays information about the total number of active translations, NAT configuration parameters, the number of addresses in the pool, and the number of addresses that have been allocated.

To verify that the NAT translation is working, it is best to clear statistics from any past translations using the **clear ip nat statistics** command before testing.
Prior to any communications with the web server, the **show ip nat statistics** command shows no current hits. After the client establishes a session with the web server, the **show ip nat statistics** command has been incremented to five hits. This verifies that the static NAT translation is taking place on R2.

```
R2# clear ip nat statistics

R2# show ip nat statistics
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/0/0
Hits: 0  Misses: 0
<output omitted>

Client PC establishes a session with the web server

R2# show ip nat statistics
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 2, occurred 00:00:14 ago
Outside interfaces:
  Serial0/1/0
Inside interfaces:
  Serial0/0/0
Hits: 5  Misses: 0

<output omitted>
```

Configuring Dynamic NAT

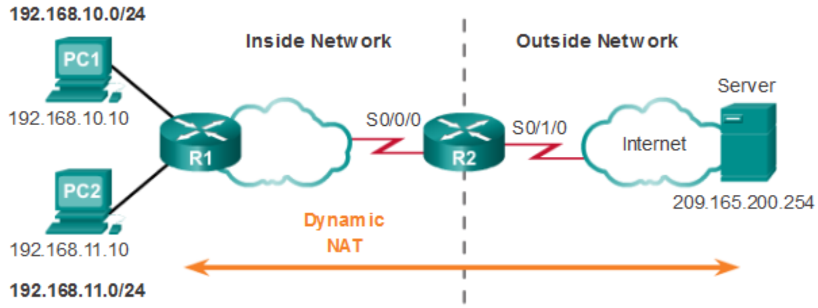
While static NAT provides a permanent mapping between an inside local address and an inside global address, dynamic NAT allows the automatic mapping of inside local addresses to inside global addresses. These inside global addresses are typically public IPv4 addresses. Dynamic NAT uses a group, or pool of public IPv4 addresses for translation.
Dynamic NAT, like static NAT, requires the configuration of the inside and outside interfaces participating in NAT.

The example topology shown below has an inside network using addresses from the RFC 1918 private address space. Attached to router R1 are two LANs, 192.168.10.0/24 and 192.168.11.0/24. Router R2, the border router, is configured for dynamic NAT using a pool of public IPv4 addresses 209.165.200.226 through 209.165.200.240.

The pool of public IPv4 addresses (inside global address pool) is available to any device on the inside network on a first-come first-served basis. With dynamic NAT, a single inside address is translated to a single outside address. With this type of translation there must be enough addresses in the pool to accommodate all the inside devices needing access to the outside network at the same time. If all of the addresses in the pool have been used, a device must wait for an available address before it can access the outside network.

Dynamic NAT

One-to-One Translation



IPv4 NAT Pool

Inside Global Address	Inside Local Address Pool
209.165.200.226	192.168.10.10
209.165.200.227	192.168.11.10
209.165.200.228	Available
...	...
209.165.200.240	Available

The steps and the commands used to configure dynamic NAT are as follows:

Step 1. Define the pool of addresses that will be used for translation using the **ip nat pool** command. This pool of addresses is typically a group of public addresses. The addresses are defined by indicating the starting IP address and the ending IP address of the pool. The **netmask** or **prefix-length** keyword indicates which address bits belong to the network and which bits belong to the host for the range of addresses.

Step 2. Configure a standard ACL to identify (permit) only those addresses that are to be translated. An ACL that is too permissive can lead to unpredictable results. Remember there is an implicit **deny all** statement at the end of each ACL.

Step 3. Bind the ACL to the pool. The **ip nat inside source list** *access-list-number* **number pool** *pool name* command is used to bind the ACL to the pool. This configuration is used by the router to identify which devices (**list**) receive which addresses (**pool**).

Step 4. Identify which interfaces are inside, in relation to NAT; that is, any interface that connects to the inside network.

Step 5. Identify which interfaces are outside, in relation to NAT; that is, any interface that connects to the outside network.

Range of address public.

public range

private range

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بتك يتم

Dynamic NAT Configuration Steps

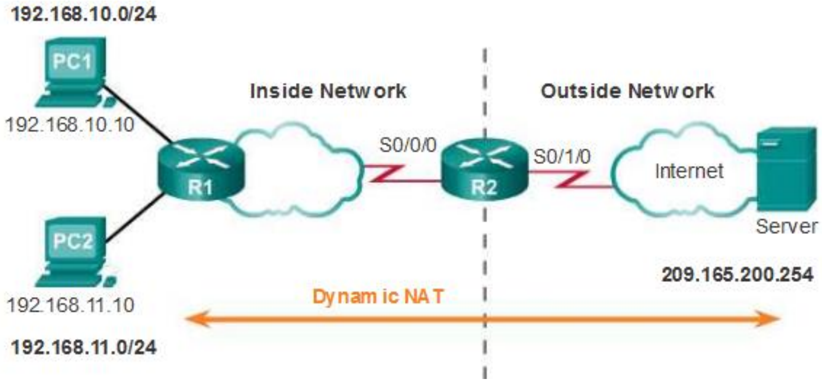
Step 1	Define a pool of global addresses to be used for translation.
	`ip nat pool name start-ip end-ip`
	`{netmask netmask` subnet mask `| ~~prefix-length prefix-length}~~`

Step 2	Configure a standard access list permitting the addresses that should be translated.
	`access-list access-list-number permit source`
	`[source-wildcard]`

Step 3	Establish dynamic source translation, specifying the access list and pool defined in prior steps.
	`ip nat inside source list access-list-number pool`
	`name`

Step 4	Identify the inside interface.
	`interface type number`
	`ip nat inside`

Step 5	Identify the outside interface.
	`interface type number`
	`ip nat outside`



Define a pool of public IPv4 addresses 209.165.200.241 to 209.165.200.250 with pool name PUBLIC-POOL.
R2(config)# ip nat pool PUBLIC-POOL 209.165.200.241 209.165.200.250 netmask 255.255.255.224
Configure ACL 2 to permit devices from 192.168.10.0/24 network to be translated by NAT.
R2(config)# access-list 2 permit 192.168.10.0 0.0.0.255
Bind PUBLIC-POOL with ACL 2.
R2(config)# ip nat inside source list 2 pool PUBLIC-POOL
Configure the proper inside NAT interface.
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
Configure the proper outside NAT interface.
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside
You successfully configured dynamic NAT.

The output of the **show ip nat translations** command shown below displays the details of the two previous NAT assignments. The command displays all static translations that have been configured and any dynamic translations that have been created by traffic.

```
R2# show ip nat translations
Pro  Inside global        Inside local    Outside local Outside global
---  209.165.200.226      192.168.10.10   ---           ---
---  209.165.200.227      192.168.11.10   ---           ---
```

By default, translation entries time out after 24 hours, unless the timers have been reconfigured with the **ip nat translation timeout** *timeout-seconds* command in global configuration mode.
To clear dynamic entries before the timeout has expired, use the clear ip nat translation global configuration mode command. It is useful to clear the dynamic entries when testing the NAT configuration.
Note:Only the dynamic translations are cleared from the table. Static translations cannot be cleared from the translation table.
the **show ip nat statistics** command displays information about the total number of active translations, NAT configuration parameters, the number of addresses in the pool, and how many of the addresses have been allocated.
Alternatively, use the **show running-config** command and look for NAT, ACL, interface, or pool commands with the required values. Examine these carefully and correct any errors discovered.

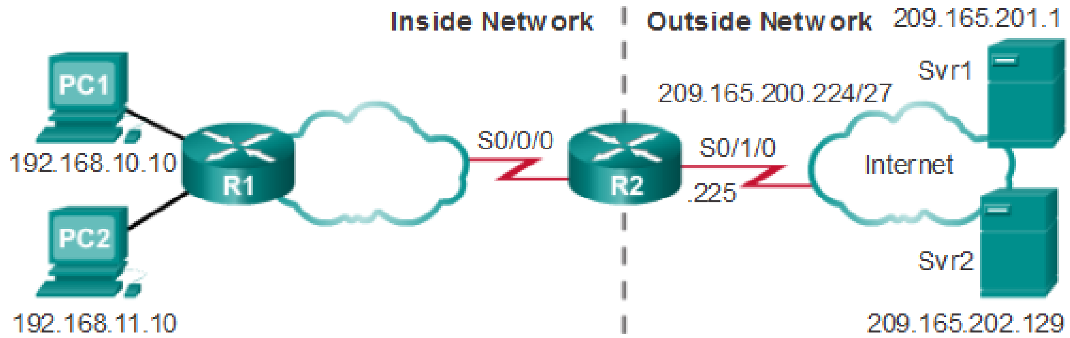
```
R2# clear ip nat statistics

PC1 and PC2 establish sessions with the server

R2# show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 0
extended)
Peak translations: 6, occurred 00:27:07 ago
Outside interfaces:
   Serial0/0/1
Inside interfaces:
   Serial0/1/0
Hits: 24 Misses: 0
CEF Translated packets: 24, CEF Punted packets: 0
Expired translations: 4
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 pool NAT-POOL1 refcount 2
  pool NAT-POOL1: netmask 255.255.255.224
start 209.165.200.226 end 209.165.200.240
type generic, total addresses 15, allocated 2 (13%), misses
0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R2#
```

Configuring Port Address Translation (PAT)

PAT (also called NAT overload) conserves addresses in the inside global address pool by allowing the router to use one inside global address for many inside local addresses. In other words, a single public IPv4 address can be used for hundreds, even thousands of internal private IPv4 addresses. When this type of translation is configured, the router maintains enough information from higher-level protocols, TCP or UDP port numbers, for example, to translate the inside global address back into the correct inside local address. When multiple inside local addresses map to one inside global address, the TCP or UDP port numbers of each inside host distinguish between the local addresses.
Note: The total number of internal addresses that can be translated to one external address could theoretically be as high as 65,536 per IP address. However, the number of internal addresses that can be assigned a single IP address is around 4,000.
There are two ways to configure PAT, depending on how the ISP allocates public IPv4 addresses. In the first instance, the ISP allocates more than one public IPv4 address to the organization, and in the other, it allocates a single public IPv4 address that is required for the organization to connect to the ISP.

Configuring PAT for a Pool of Public IP Addresses

If a site has been issued more than one public IPv4 address, these addresses can be part of a pool that is used by PAT. This is similar to dynamic NAT, except that there are not enough public addresses for a one-to-one mapping of inside to outside addresses. The small pool of addresses is shared among a larger number of devices. The primary difference between this configuration and the configuration for dynamic, one-to-one NAT is that the **overload** keyword is used. The **overload** keyword enables PAT.
The example configuration shown below establishes overload translation for the NAT pool named NAT-POOL2. NAT-POOL2 contains addresses 209.165.200.226 to 209.165.200.240. Hosts in the 192.168.0.0/16 network are subject to translation. The S0/0/0 interface is identified as an inside interface and the S0/1/0 interface is identified as an outside interface.



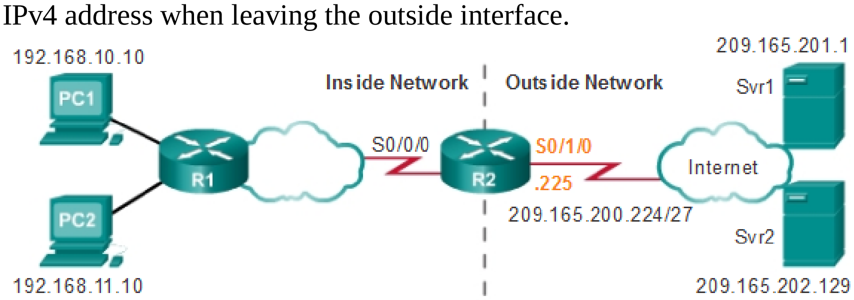
R2(config)# ip nat pool NAT-POOL2 209.165.200.226 209.165.200.240 netmask 255.255.255.224
R2(config)# access-list 1permit 192.168.0.0 0.0.255.255
R2(config)# ip nat inside source list 2 pool NAT-POOL2 overload
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside

كل واحد صار له ال public addr بتعرف تميزهم بال port .

Configuring PAT for a Single Public IPv4 Address

The following figure shows the topology of a PAT implementation for a single public IPv4 address translation. In the example, all hosts from network 192.168.0.0/16 (matching ACL 1) that send traffic through router R2 to the Internet will be translated to IPv4 address 209.165.200.225 (IPv4 address of interface S0/1/0). The traffic flows will be identified by port numbers in the NAT table, because the **overload** keyword was used.

With only a single public IPv4 address available, the overload configuration typically assigns the public address to the outside interface that connects to the ISP. All inside addresses are translated to the single IPv4 address when leaving the outside interface.



NAT Table

Inside Global Address	Inside Local Address	Outside Local Address	Outside Global Address
209.165.200.225:1444	192.168.10.10:1444	209.165.201.1:80	209.165.201.1:80
209.165.200.225:1445	192.168.10.11:1444	209.165.202.129:80	209.165.202.129:80

R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R2(config)# ip nat source list 1 interface serial 0/1/0 overload
R2(config)# interface serial0/0/0
R2(config-if)# ip nat inside
R2(config)# interface serial0/1/0
R2(config-if)# ip nat outside

The configuration is similar to dynamic NAT, except that instead of a pool of addresses, the **interface** keyword is used to identify the outside IPv4 address. Therefore, no NAT pool is defined.

The same commands used to verify static and dynamic NAT are used to verify PAT. The **show ip nat translations** command displays the translations from two different hosts to different web servers. Notice that two different inside hosts are allocated the same IPv4 address of 209.165.200.226 (inside global address). The source port numbers in the NAT table differentiate the two transactions.

```
R2# show ip nat translations
Pro Inside global          Inside local          Outside local
tcp 209.165.200.226:51839  192.168.10.10:51839   209.165.201.1:80
tcp 209.165.200.226:42558  192.168.11.10:42558   209.165.202.129:8
R2#
```

The **show ip nat statistics** command verifies that NAT-POOL2 has allocated a single address for both translations. Included in the output is information about the number and type of active translations, NAT configuration parameters, the number of addresses in the pool, and how many have been allocated.

```
R2# clear ip nat statistics

R2# show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 2 extended)
Peak translations: 2, occurred 00:00:05 ago
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/1/0
Hits: 4  Misses: 0
CEF Translated packets: 4, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 3] access-list 1 pool NAT-POOL2 refcount 2
 pool NAT-POOL2: netmask 255.255.255.224
     start 209.165.200.226 end 209.165.200.240
     type generic, total addresses 15, allocated 1 (6%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R2#
```

Procedures:

Dear students, please note that the lab problems sheet, the packet tracer activities and the practical discussion videos have been uploaded on your Microsoft Teams group. You are required to carefully study this experiment and then complete the lab sheet.

References

Enterprise Networking, Security, and Automation - Cisco Networking Academy
https://www.netacad.com